

Lab W09: Sort It Yourself

Data Structures, Grade Sheet

Name:

Date:

Language used (circle one): Java C/C++

Submission and setup

3 pts

Submitted the entire labw09 project (Eclipse project for Java, full folder including the Makefile for C/C++), not just individual files

/ 1

Project compiles and runs without errors

/ 1

Folder and package layout left as given, not renamed or rearranged

/ 1

Code quality

3 pts

Naming and formatting are reasonable and consistent with the starter code

/ 1

No leftover dead code, stray debugging output, or commented-out blocks

/ 1

reverse and beltSort follow the existing style and generic bound used by the other methods in Sorts

/ 1

Development Process

13 pts

Comment at the top of beltSort explains the strategy in the student's own words (three or four sentences)

/ 2

Photos of handwritten development submitted, worked out before the code

/ 1

Trace is annotated: each step names its reverse(k) call, the value of k, and the resulting row

/ 1

Notes are clear and thorough to follow the reasoning, not just pseudocode or code

/ 9

Required Behavior

10 pts

reverse(k) reverses theArray[0..k-1] in place

/ 2

reverse(k) leaves every item from index k onward untouched

/ 2

beltSort changes the order only through reverse(k); no direct swaps or other rearranging

/ 2

Sorter demonstrates beltSort on the Integer array, below the marked point

/ 2

Sorter demonstrates beltSort on the Droid array, below the marked point

/ 2

Lab W09: Sort It Yourself (continued)

Test Cases

11 pts

Empty array (0 items)		/ 1
One item		/ 1
Two items, already in ascending order		/ 1
Two items, in descending order		/ 1
More than two items, already sorted in ascending order		/ 1
More than two items, already sorted in descending order		/ 1
More than two items, all equal		/ 1
More than two items, random order with duplicate values		/ 1
More than two items, largest already at the front (for example 9, 1, 2, 3)		/ 1
Droid array sorts correctly by ID		/ 1
Outputs the results the way Sorter is already coded to do		/ 1

Total

/ 40

Notes